

Technology for Success: Computer Concepts

Critical Thinking Challenges Mentoring Guide

WHAT IS IT FOR? Explains each possible outcome of various decisions/paths that a student could take. Prescribes a strategy and recommended guidance that an instructor can give a student based on the specific decision path chosen.

RECOMMENDED USE: Instructors can view (1) high-level tips by module and (2) strategies for each step in the CTC to guide students on how to choose the optimal path. The Guide can be used as instructors are reviewing results and/or as students are completing the CTCs, possibly as an in-class activity. It can aid the instructor in determining where in the path the student deviated from the optimal path and how to help the student get on track.

Note: Turn on your Navigation Pane on the View tab in Word to see headings for easy navigation in this document.

High-level tips by module

All modules
As in the real world, there are many ways to complete the tasks, options for fixing mistakes along the way, and some outcomes that are more preferable than others. When you finish, decide whether the outcome is optimal. Then try the scenario again to see if you can get a different outcome!
Module 1
This scenario gives you the opportunity to decide whether to use technology throughout your day. At each step, think carefully about why you might use technology, and how it could benefit you.
Look for ways to use modern technology to streamline your tasks and help you get to class on time.
Often, newer technology is developed to improve efficiency with everyday tasks.
Module 2
This scenario gives you the opportunity to practice working with different ways of finding information. At each step, think carefully about which way would be best to search for information and how to customize your search to obtain meaningful results.
As you think about how to find information for this new software, consider which information sources will provide you with the most reliable and relevant information.
Try to balance the need for urgency with the need for accuracy.
Make sure you accurately understand the user needs you're trying to fulfill.
Module 3

This scenario gives you the opportunity to decide what kind of computer is most appropriate for the scenario. At each step, think carefully about what hardware and features are needed most while staying within your budget.

Think carefully about what kind of system will best meet the needs of the user.

Consider which information sources will provide you with the most reliable and relevant information.

Make sure that both hardware and software choices are appropriate to the demands of the work and the need for portability while trying to stay within budget.

Consider what tradeoffs you can reasonably make with the new computer to be able to also purchase a printer.

Module 4

This scenario gives you the opportunity to practice working with different versions of a file. At each step, think carefully about which file you need, whether that file exists online or on your local computer, and what other versions of the file other people might be able to see at any given time.

Consider which tool will do what you need it to do.

Think very carefully about where a file is located—both its folder location and its device location.

Try to choose the right operation to minimize the chances of mistakes or misunderstandings.

Make sure you're communicating accurate information to others in the scenario.

Module 5

This scenario gives you the opportunity to practice working with different kinds of applications. At each step, think carefully about which app will best accomplish your goals, and what features within the app will perform the tasks you need.

Be careful to choose the best tool for the job at each step along the way.

Remember to take your time in forming your proposal so it has the best chance of going over well with the boss.

Consider the best sources of information on which to build your proposal.

Make sure you present your findings in a clean, professional, and efficient manner.

Module 6

This scenario gives you the opportunity to make several decisions to try to keep your data and your client's data safe. At each step, think carefully about how security might be compromised and find ways to avoid unnecessary risks.

Think about how to protect data when it is stored so that unauthorized users can't get to it.

Carefully consider what kind of wireless network is safe enough to use for transmitting sensitive information.

Be careful about what kinds of emails you open, and what you do with those emails.

Module 7

This scenario gives you the opportunity to decide what kinds of audio, video, and photo files will work best in certain situations. At each step, think carefully about quality versus convenience and decide which of these priorities is most important.

Remember to consider your budget along the way.

Keep in mind what devices will be used to access various multi-media files and which file type is best for that kind of device.

Consider licensing restrictions.

Think about how to maximize accessibility to as many people as possible.

Module 8

This scenario gives you the opportunity to decide how best to design an app to attract customers to your store. At each step, think carefully about what app users most want in an app and how to best connect with as many potential customers as possible.

Consider the different user expectations for free apps versus paid apps.

Find ways to keep your customers coming back to your new app.

Think about which job role does what in developing an app.

Module 9

This scenario gives you the opportunity to decide what kinds of changes are appropriate for a website update. At each step, think carefully about how best to make professional looking changes.

Consider what style is most appropriate for this particular company.

Consider what tools will work best for the job to accomplish your goals.

Remember to consider the budget along the way.

Think about the technical side, such as HTML tags and components of the website design.

Module 10

This scenario gives you the opportunity to make decisions on how best to configure a network. At each step, think carefully about what configurations will best serve the customer's needs.

Consider what hardware will provide needed performance without costing more than necessary.

Think about how to section off parts of the network to protect private data.

Find ways to connect parts of the network that need to communicate with each other.

Keep in mind the importance of good security habits.

Module 11

This scenario gives you the opportunity to practice working with different ways of finding information. At each step, think carefully about which way would be best to search for information and how to customize your search to obtain meaningful results.

Keep your target audience in mind as you make decisions about how to market your new product.

Use modern marketing techniques that take advantage of what social media has to offer.

Remember to protect the privacy of your customers.

Module 12

This scenario gives you the opportunity to make decisions about an upgrade to a business website used to sell products to other businesses. At each step, think carefully about what kinds of design changes are best for the business processes the website supports.

Consider how to use modern technology to make everyone's work easier and more efficient.

Look for ways to save money by using pay-as-you-go services.

Choose upgrades to the website that support existing business processes and relationships rather than requiring those things to change to fit the website.

Module 13

This scenario gives you the opportunity to make decisions about how to design a database table and run a query on the new table. At each step, think carefully about what kind of data should be entered in each field so you can configure the field appropriately.

Consider which field will include unique information about each client so it can serve as the table's primary key.

Think about what kinds of controls you can build into the table to help users enter accurate data.

Keep in mind that each employee of a company typically works with many clients.

Module 14

This scenario gives you the opportunity to make ethical decisions on the job. At each step, think carefully about which option shows a sense of personal responsibility and accountability to others.

Consider what laws, rules, or guidelines might apply in each situation.

Think about how your actions might affect other people, even if they're not aware of your decisions.

Imagine how you would feel if someone treated you the way you're treating another person.

Play-by-play strategies

Module 1

Orientation		In this activity, students are challenged to choose how to complete typical daily tasks. Along the way, they'll need to decide whether to use modern technology or do things the "old-fashioned way." An important point to remember here is technology saves time and helps students meet more of their goals. Students will also need to resolve unexpected problems. Students should prioritize using technology to do tasks more quickly.		
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.	
Level 1	Decision 1	Make coffee	Students are given the opportunity to use IoT technology to start a pot of coffee. Optimally, students will choose to use Alexa and the smart plug. "The goal here is to use technology as much as possible to help you achieve all your goals today in a timely manner. What's the fastest way to get that pot of coffee running?"	
Level 2	Decision 1	Pack bag	While the traditional way of doing things might be more comfortable, technology can really help people to work more efficiently. Optimally, students will choose to bring the laptop and use OneNote during class. "How can technology benefit you during class?"	
	Decision 2	Pack bag	While the traditional way of doing things might be more comfortable, technology can really help people to work more efficiently. Optimally, students will choose to bring the laptop and use OneNote during class. "How can technology benefit you during class?"	
Level 3	Decision 1	Check weather	A personal assistant such as Alexa is designed to make information access quick and efficient. Optimally, students will choose to ask Alexa for the weather instead of waiting on the TV's broadcast. "Which option will help you get the day's weather forecast the fastest?"	
	Decision 2	Check weather	A personal assistant such as Alexa is designed to make information access quick and efficient. Optimally, students will choose to ask Alexa for the weather instead of waiting on the TV's broadcast. "Which option will help you get the day's weather forecast the fastest?"	
Level 4	Decision 1	Bank deposit	The security of online banking has come a long way and can now be a reliable means of making deposits. Optimally, students will choose to make the deposit online. "You've got to get this check in the bank so you can pay rent, but you also need to get to class on time. Which option will best help you meet both goals?"	

	Decision 2	Bank deposit	The security of online banking has come a long way and can now be a reliable means of making deposits. Optimally, students will choose to make the deposit online. “You’ve got to get this check in the bank so you can pay rent, but you also need to get to class on time. Which option will best help you meet both goals?”
	Decision 3	Bank deposit	The security of online banking has come a long way and can now be a reliable means of making deposits. Optimally, students will choose to make the deposit online. “You’ve got to get this check in the bank so you can pay rent, but you also need to get to class on time. Which option will best help you meet both goals?”
Level 5	Decision 1	Travel method	On this track, students chose to make a deposit at the bank. Unfortunately, their car won’t start and they’ll need a ride to get to the bank. Optimally, students will choose to use Lyft. “Unfortunately, your car won’t start. As you consider your options, remember that technology can often help save time.”
Level 6	Decision 1	Pay rent	On this track, students have lost significant amounts of time. Optimally, they will choose to pay rent using the Zelle app and hopefully get back some lost time. “You’re really running behind now. What’s the fastest way to pay your rent so you can get to class?”
	Decision 2	Pay rent	On this track, students have made good decisions that are saving them time. Optimally, they will choose to pay rent using the Zelle app and hopefully stay on track to make it to class on time. “You’ve made some great decisions so far, and you’re running early. This is a good time to keep using technology to your advantage.”
Level 7	Decision 1	Get books	Here, students can face the crowds at the bookstore or get their books online. Optimally, they’ll choose the online option so they don’t miss class entirely. “You’ve got to make up for some lost time. How can technology help?”
	Decision 2	Get books	Here, students can face the crowds at the bookstore or get their books online. Optimally, they’ll choose the online option so they won’t be late to class. “Technology hasn’t failed you yet. Which option will get you to class the fastest?”
Outcome 1 [Sub-optimal]		Late to class	For students who reach Outcome 1: “As you try again, think about ways you can be more efficient with your time by using technology. Even though technology may be challenging to learn how to use, it can also save you a great deal of time throughout your day.”
Outcome 2 [Optimal]		Early to class	For students who reach Outcome 2: “Great job relying on technology to help you use your time efficiently.”

Outcome 3
[Dead end]

Missed
class

For students who reach Outcome 3:

“As you try again, think about ways you can be more efficient with your time by using technology. Even though technology may be challenging to learn how to use, it can also save you a great deal of time throughout your day.”

Module 2

Orientation		In this activity, students are challenged to use good research methodology in gathering information about some software for their company. Along the way, they'll need to decide which resources to use for finding information, whether to keep digging deeper for more information, and at what point they're sufficiently informed to make a solid recommendation. An important point to remember here is to carefully evaluate the source of information to determine if that source is credible, accurate, reasonable, and supported by outside resources. Students will also need to balance information from the Internet with information from people in the scenario. Students should prioritize reliable sources and ensure they have completed a thorough review of the options before making their recommendations.	
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.
Level 1	Decision 1	Info source	Students are asked to choose between searching online for initial information or consulting with a salesperson at a local store. Either option can be a good place to start. "The goal here is to find a trustworthy source of information that will help you make your recommendations."
Level 2	Decision 1	Buy or not	Students are tempted with the opportunity to make a purchase decision right away. Optimally, students will wait until they have collected more thorough information. "Consider whether you have enough information now to make a purchasing decision."
Level 3	Decision 1	Info source	At this point, students made a poor decision to buy prematurely. The product didn't work out well. Students are given another opportunity to research online, or they can consult with colleagues to determine user needs. Either option will provide needed information. "The goal here is to find a trustworthy source of information that will help you make your recommendations."
Level 4	Decision 1	Choose link	Using a search engine typically provides an overwhelming amount of information. Ideally, students would have a good idea of what information they're looking for and target their search efforts more carefully. In this case, neither option relies on much pre-existing knowledge about the search material. "It's hard to tell which links might be useful. What is a good way to start exploring your search results?"
Level 5	Decision 1	Info source	The students learn by trial and error that some search results are actually advertisements. Students are given three options to refine their search, any of which will provide needed information. "You didn't get the information you needed. What else can you try that will give you better quality information?"

	Decision 2	Info source	The students learn by trial and error that clicking a random link is not an efficient way to use search results. Students are given two options to refine their search efforts, either of which will provide needed information. “You didn’t get the information you needed. What else can you try that will give you better quality information?”
Level 6	Decision 1	Confirm online or contact sales	As students progress in their research, their search efforts should become more refined and targeted. Students are given options on how to redirect their search. Optimally, students will perform a targeted search online for specific, unbiased information. “You’re getting some good ideas to build on. Where can you get reliable background information?”
	Decision 2	Check site or contact companies	As students progress in their research, their search efforts should become more refined and targeted. Students are given options on how to redirect their search. Optimally, students will perform a targeted search online for specific, unbiased information. “You’re getting some good ideas to build on. Where can you get reliable background information?”
Level 7	Decision 1	More info	As the search continues, students need to mentally check back with the original purpose of their research: to meet user needs. Optimally, students will consult with the intended users before making any software recommendations. “Opinions and facts are both important when researching a purchase, although it’s important to consider each person’s bias and the relevance of their opinions and preferences. Whose opinion is most relevant to this decision?”
Level 8	Decision 1	Feedback	As the search continues, students need to mentally check back with the original purpose of their research: to meet user needs. Optimally, students will consult with the intended users before making any software recommendations. “Consider whether you have enough information now to make a purchasing decision.”
Level 9	Decision 1	Timing	It can be tempting, when comparison shopping, to make a purchasing decision prematurely, especially when having to wait on other people for additional information. In this case, because the software will be used by so many different people with different needs, the student will optimally choose to wait on survey results. “Do you think feedback from the people who will use the product is important enough to wait for before making your recommendations?”
Level 10	Decision 1	Stay late?	Even though the survey results are ambiguous, taking the time to sort through the results and understand user needs will inform the recommendation. When making recommendations to decision makers, it’s important to provide them with all relevant information and be prepared to answer any questions they might have. Optimally, students will choose to stay late and thoroughly analyze survey results.

			“Even though the survey results are mixed, is it worth it to dig through those results for additional insights? Remember, you’re new to this company and need to make a good impression.”
Outcome 1 [Sub-optimal]	No clear recommendations	For students who reach Outcome 1: “As you try again, think about ways you can find more reliable information that isn’t as biased and that connects with the concerns and needs of the people who will be using this system.”	
Outcome 2 [Optimal]	Two excellent options	For students who reach Outcome 2: “Great job finding relevant and insightful information—this whole project is helping you better understand your new company while also making a good impression.”	

Module 3

Orientation		In this activity, students must choose what kind of computer will work best for schoolwork while staying within a given budget. Along the way, they'll be tempted with quick and cheap options or with high-end hardware. It's important for students to keep the end goal in mind: a portable computer that can support Office and doesn't bust the budget. Ideally, students will have enough money left in the budget to also buy a small printer. Students should prioritize portability, affordability, and the capability of supporting Microsoft Office, which is required for their schoolwork.	
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.
Level 1	Decision 1	Choose computer type	Students are asked to choose a device that is portable and can support Microsoft Office. The only option that meets these two requirements is a laptop. "The goal here is to choose a mobile device that you can install Microsoft Office on."
Level 2	Decision 1	Choose info source	After choosing a device type, students must collect some initial information on the available options. It's a little too soon to be reading reviews and looking for a good deal. Optimally, students will go to the store to compare available models and get a better idea of what they need. "Are you ready to start looking for a good deal, or do you need more information first?"
	Decision 2	Choose info source	Students are tempted with a quick and easy option: get a computer just like Colin's because he's happy with his decision. Optimally, students will do more research either by reading reviews of top laptop models or by visiting a store to compare some laptop models. "Colin likes his Chromebook. But is that the right decision for your situation?"
Level 3	Decision 1	Choose purpose	After examining several models, students must choose a set of priorities in order to narrow down the options. Optimally, students will choose to look for a laptop suited for schoolwork. "Keeping in mind that you also want to be able to get a printer, is it better to put a lot of money into a gaming laptop or better to focus on what you need for school?"
Level 4	Decision 1	Choose screen size, weight	Here, students chose to look for a gaming laptop. However, they still must comply with a reasonable budget. Optimally, students will choose the smaller screen size in order to ensure the laptop is more portable and not quite as expensive. "A gaming laptop can be a lot fun, but you still have to make sure the laptop will meet your needs for school. Which is better for school: a larger and heavier laptop or a smaller and more portable one?"
	Decision 2	Choose screen size, weight	Here, students chose to get a budget-friendly laptop for school. This gives them more options in available sizes to choose from, depending on their preferences. Still, a large, 17-inch laptop is too big to easily stick in a backpack and carry around for school. Optimally, students will choose either a 13-inch or 15-inch laptop.

			"You've got a lot of size options at this price point. Which is better for school: a larger and heavier laptop or a smaller and more portable one?"
Level 5	Decision 1	Choose OS	Now that students have selected a smaller laptop size, they need to choose an operating system. Microsoft Office, which is required for school, can't be installed on a Chromebook. And macOS is too expensive to fit into their current budget. Optimally, students will choose Windows. "Chromebooks are pretty cheap, but will a Chromebook do what you need it to do for school?"
	Decision 2	Choose OS	Now that students have selected a laptop size, they need to choose an operating system. Microsoft Office, which is required for school, can't be installed on a Chromebook. Optimally, students will choose either Windows (which leaves more room in the budget for a printer) or macOS (which is easily compatible with an iPhone). "Chromebooks are pretty cheap, but will a Chromebook do what you need it to do for school?"
Level 6	Decision 1	Choose performance	Students are given another opportunity to decide how much money to invest in the computer in order to meet the needs given in the scenario. Optimally, students will choose lower end hardware and hopefully have enough money left over for a printer. "Is it better to max out your budget on the computer or hold back in the hope of being able to also get a printer?"
	Decision 2	Choose new or refurbished	Unfortunately, a Mac computer is more expensive than a Windows computer. The only way to stay within the given budget is to buy a refurbished Mac computer. "Mac laptops are pretty expensive—your scholarship isn't enough to pay for a new one. Is it better to pay some more money out of pocket for a new one, or buy a certified refurbished Mac so you can stay in your budget?"
Level 7	Decision 1	Choose brand	Brand-name computers with these specifications cost more than the allotted budget from the scholarship money. The only way to stay within the given budget is to buy an off-brand laptop. "Brand-name laptops are pretty expensive—your scholarship isn't enough for one of these highly rated systems. Is it better to pay some more money out of pocket or buy an off-brand laptop so you can stay in your budget?"
	Decision 2	Choose storage	Students on this path are trying to make good decisions to stay within their budget. Here, students face yet another decision that could risk their budget: how much storage space to buy. Optimally, students will choose the smaller SSD. "SSDs are fast, but it's also kind of expensive to get one with a lot of storage space. How important is having all that storage?"
	Decision 3	Choose performance	Even a refurbished Mac can be expensive if it comes with some high-end hardware. One way to keep the price down is to get less RAM.

			“Having a lot of RAM is really nice, but it’s also expensive. How important is it to stay within your budget?”
Outcome 1 [Dead end]	Device does not meet the requirements of the scenario	For students who reach Outcome 1:	“As you try again, think about what requirements this computer must be able to meet: it must be easily portable so you can take it to class, and you must be able to install Office on it.”
Outcome 2 [Optimal]	Good laptop within budget and a printer	For students who reach Outcome 2:	“Good work finding a computer within your budget that also does everything you need it to do!”
Outcome 3 [Sub-optimal]	Overbudget and can’t get a printer	For students who reach Outcome 3:	“As you try again, consider ways to stay within your budget so you can also buy a printer. Think carefully about the size, operating system, brand, and amount of storage space in order to be able to have everything you need to be successful at school.”

Module 4

Orientation		In this activity, students must decide how best to make updates to a flyer document. Along the way, they'll need to track where the document is located and who has access to it. It's important for students to make appropriate changes to the file's location and type. Ideally, students will present the expected final version to the appropriate contact at the other company. Students should prioritize accuracy and clear communication.	
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.
Level 1	Decision 1	Choose OS	Students are asked to choose an operating system to use for making changes to this file. While either OS will work, students will need to keep in mind what tools are available to them in each operating system. "Your choice here will affect the options available to you in later steps. Is one OS better than the other for this project?"
Level 2	Decision 1	Choose app	After choosing an OS, students must decide which utility will give them access to the needed file in OneDrive. Of these two options, only the Firefox browser can access the OneDrive folder. "Which of these tools will let you access OneDrive?"
	Decision 2	Choose app	After choosing an OS, students must decide which utility will give them access to the needed file in OneDrive. Either of these options will work. "Which of these tools will let you access OneDrive?"
Level 3	Decision 1	Choose file path	Nautilus can't access OneDrive, so there's no option here that will give students access to the correct file. Students can still earn points for good decisions in later steps. They will have the opportunity to benefit from a redemption step later. "Where do you think that OneDrive file is located?"
	Decision 2	Choose file path	Now that they're on the OneDrive website, students need to find the correct file. The Files folder leads to the correct AtwaterFiles folder in OneDrive, while the other file path leads to files stored in the local computer's Downloads folder. "Think carefully about where the needed file is located. Which file path will get you there?"
	Decision 3	Choose file path	Students on this track are using File Explorer to access OneDrive. The correct file is in the AtwaterEvents folder on OneDrive, not the AtwaterEvents folder on the local PC. "Think carefully about where the needed file is located. Which file path will get you there?"
Level 4	Decision 1	Choose cut or copy	By cutting and pasting, the student removes the file from its shared location on OneDrive so no one else will take it to make changes.

			“How can you ensure that no one else will make changes to the file while you’re working on it?”
	Decision 2	Choose cut or copy	By cutting and pasting, the student removes the file from its shared location on OneDrive so no one else will take it to make changes. “How can you ensure that no one else will make changes to the file while you’re working on it?”
Level 5	Decision 1	Choose save or save as	Only the Save As function will allow the student to save the file as a PDF, which is a different file type than the what the current file has. “How can you change the file type?”
	Decision 2	Choose save or save as	Only the Save As function will allow the student to save the file as a PDF, which is a different file type than the what the current file has. “How can you change the file type?”
	Decision 3	Choose save or save as	Only the Save As function will allow the student to save the file as a PDF, which is a different file type than the what the current file has. “How can you change the file type?”
Level 6	Decision 1	Choose save location	The new PDF file needs to be saved where other people can get to it. OneDrive is the only option here that will work. “How can you ensure that other people can get to this file?”
Level 7	Decision 1	Choose message to Liam	If earlier steps have been performed correctly, the file should now be saved to OneDrive where Liam can get to it. The email should include this information. However, on this track, the student has made too many mistakes for Liam to have access to the correct file. “What do you want Liam to know about the file’s location?”
	Decision 2	Choose message to Liam	If earlier steps have been performed correctly, the file should now be saved to OneDrive where Liam can get to it. The email should include this information. “What do you want Liam to know about the file’s location?”
Outcome 1 [Sub-optimal]	Liam doesn’t get the file he needs with the required changes		For students who reach Outcome 1: “As you try again, think about where the master file is located and how to get to it. Carefully choose which application can access that location, and be careful to track the file’s location as you move it around. You also need to change the file type, which can’t be done with a simple Save operation.”
Outcome 2 [Optimal]	The flyer looks great!		For students who reach Outcome 2: “Nice work! Liam got the file he needs, and it has all the correct updates.”

Module 5

Orientation		In this activity, students are challenged to collect useful data in a way that will support a proposed change at work. Along the way, they'll need to decide how best to get the most insightful data, how to present that data in the most impactful way, and how to communicate clearly with others. An important point to remember here is choose the right tool for the job in each step. Students will also need to keep in mind good business practices for professional communication. Students should prioritize professionalism, clear communication, and choosing good data to support their proposal.	
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.
Level 1	Decision 1	Choose app	Students need to use an authoritative source of information to support their proposed schedule changes. The best option here is to go directly to the data itself, which is held in the Access database. "The goal here is to use an authoritative source of information. Where can you get the data you need to make your case for the schedule changes?"
Level 2	Decision 1	Choose device	These kinds of changes will require a significant shift in business processes. To justify such a change, the proposal itself needs to be taken seriously by the person proposing it. To do this, the student needs supporting data before presenting the proposal to the boss. "What can you do here to help make sure your proposal is taken seriously?"
	Decision 2	Choose tool	In Access, a filter can refine the data shown in a single table. However, the kind of data needed for the proposal relies on more complex relationships between data. Therefore, a query is the best option here. "Which Access tool will help you find data that thoroughly addresses the question you've posed?"
Level 3	Decision 1	Choose app	A spreadsheet application can intuitively interpret data in charts and graphs. This will give a more accurate representation of the data. "Which of these applications can easily convert numbers into visual representations of that data?"
	Decision 2	Choose app	A spreadsheet application can intuitively interpret data in charts and graphs. This will give a more accurate representation of the data. "Which of these applications can easily convert numbers into visual representations of that data?"
	Decision 3	Choose app	A spreadsheet application can intuitively interpret data in charts and graphs. This will give a more accurate representation of the data. "Which of these applications can easily convert numbers into visual representations of that data?"
Level 4	Decision 1	Choose app	Both Excel and Sheets can do the work needed here.

			“Which of these apps would you rather work with to do the job?”
	Decision 2	Choose native/web app	Both Paint and draw.io can do the work needed here. “Which of these apps would you rather work with to do the job?”
	Decision 3	Choose app	Both Excel and Sheets can do the work needed here. “Which of these apps would you rather work with to do the job?”
	Decision 4	Choose native/web app	While both Paint and draw.io can do the work needed here, the underlying data on this track is not reliable. “Which of these apps would you rather work with to do the job?”
Level 5	Decision 1	Choose chart type	A column chart compares categories of data with one another, such as planned absences on each weekday. “Which chart type will best show a comparison of data for each weekday?”
	Decision 2	Choose file type	A JPEG shows a static image while a GIF typically shows a short animation. In this case, the JPEG is a better fit. “Which of these image types will give a professional, clean look for the presentation and make it easy to understand the data?”
	Decision 3	Choose chart type	A column chart compares categories of data with one another, such as planned absences on each weekday. “Which chart type will best show a comparison of data for each weekday?”
	Decision 4	Choose file type	A JPEG shows a static image while a GIF typically shows a short animation. In this case, the JPEG is a better fit. “Which of these image types will give a professional, clean look for the presentation and make it easy to understand the data?”
Level 6	Decision 1	Choose font	For a professional presentation, a clean font is preferred. “Which type of font will make the best impression for your presentation?”
	Decision 2	Choose font	For a professional presentation, a clean font is preferred. “Which type of font will make the best impression for your presentation?”
Level 7	Decision 1	Choose view	The Slide Show view will show one slide at a time on the whole screen, which makes it easy to spot errors or alignment problems. “Which view will help you check each slide carefully?”
	Decision 2	Choose view	The Slide Show view will show one slide at a time on the whole screen, which makes it easy to spot errors or alignment problems.

			“Which view will help you check each slide carefully?”
Level 8	Decision 1	Choose share	The boss won’t be able to access her inbox on her desk until the next day. Both email and OneDrive allow for electronic delivery of the file that she can access from home. “Which of these options will get the file to your boss while she’s at home?”
	Decision 2	Choose share	The boss won’t be able to access her inbox on her desk until the next day. Both email and OneDrive allow for electronic delivery of the file that she can access from home. “Which of these options will get the file to your boss while she’s at home?”
Outcome 1 [Sub-optimal]		Boss vetoes the proposal	For students who reach Outcome 1: “As you try again, think about how to get the most relevant data, how best to present that data in a clean and professional manner, and what tool you need at each step to help you do that step well.”
Outcome 2 [Optimal]		Boss likes the proposal and invites the CEO to the meeting	For students who reach Outcome 2: “Your boss likes what she’s seeing so far and has invited the CEO to the meeting. Nice work!”

Module 6

Orientation		In this activity, students are challenged to keep data safe while using a new laptop during a client meeting. Along the way, they'll need to decide what network connections are safe and what techniques help to protect data. An important point to remember here is the easiest way to do something typically is not the safest way. Students will also need to choose how to keep themselves safe when using email. Students should prioritize safety over convenience.	
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.
Level 1	Decision 1	Choose where to save passwords	Students are tempted with the easy option to save passwords in a browser. However, browsers typically store passwords in plaintext, which is not safe. A password manager is a much more secure option. Note that the division of tracks here differentiates between which students have safely stored their passwords in a password manager, and which students have chosen the less secure alternative. This decision will affect the possible outcomes available to students. "The goal here is to find a safe place to store your passwords. Which of these is the more secure option?"
Level 2	Decision 1	Choose focus	While it might not seem like a security issue, staying focused at work and keeping a clear mind can affect the kinds of decisions people can make. Here, students are tempted with distracting themselves by playing on their phones or thinking over the meeting's agenda to be prepared for whatever direction the discussion takes and focus on building a trust relationship with the client. "Every decision you make can influence security issues. How can you be best prepared to start building a trust relationship with your client?"
	Decision 2	Choose focus	While it might not seem like a security issue, staying focused at work and keeping a clear mind can affect the kinds of decisions people can make. Here, students are tempted with distracting themselves by playing on their phones or thinking over the meeting's agenda to be prepared for whatever direction the discussion takes and focus on building a trust relationship with the client. "Every decision you make can influence security issues. How can you be best prepared to start building a trust relationship with your client?"
Level 3	Decision 1	Choose Wi-Fi	When using a free Wi-Fi hotspot, it's important to choose one offered by a reputable business. The store's own SpaceCups Wi-Fi is most likely safer than random Wi-Fi access offered by someone else. "Not every free Wi-Fi hotspot is safe. How can you increase your chances of connecting to a safe network?"
	Decision 2	Choose Wi-Fi	Here, students are given the option to ask the manager to borrow their private Wi-Fi for more secure access to transmit business information. Not every manager would agree to this, but it doesn't hurt to ask if you have a legitimate reason for concern.

			"You know you need to transmit sensitive data across the Internet. How can you increase your chances of connecting to a safe network?"
Level 4	Decision 1	Choose email	Students are given the option to start with reading a friend's email or checking an email seeming to come from their bank. At this point, they don't have enough information to determine whether either of these emails is problematic. However, on this track, the student's laptop is already compromised because of using an unsafe network connection. "Which email would you rather start with?"
	Decision 2	Choose email	Students are given the option to start with reading a friend's email or checking an email seeming to come from their bank. At this point, they don't have enough information to determine whether either of these emails is problematic. "Which email would you rather start with?"
	Decision 3	Choose email	Students are given the option to start with reading a friend's email or checking an email seeming to come from their bank. At this point, they don't have enough information to determine whether either of these emails is problematic. "Which email would you rather start with?"
Level 5	Decision 1	Choose action	The friend's email obviously contains clickbait. Optimally, the student will delete the email. "Do you think this link is legitimate?"
	Decision 2	Choose action	The bank email could be legitimate, but it's never a good idea to click a link in an email. It's better to use a browser to navigate directly to the bank's website and sign in that way, then look for indications that the alert is valid. "Do you think this link is legitimate?"
	Decision 3	Choose action	The friend's email obviously contains clickbait. Optimally, the student will delete the email. "Do you think this link is legitimate?"
	Decision 4	Choose action	The bank email could be legitimate, but it's never a good idea to click a link in an email. It's better to use a browser to navigate directly to the bank's website and sign in that way, then look for indications that the alert is valid. "Do you think this link is legitimate?"
Level 6	Decision 1	Choose method	Even with using a store's Wi-Fi hotspot, there's no guarantee it's a safe network. Optimally, students will choose to return to the office to upload the report over a secure connection. "What's the best way to protect the sensitive data included in this report?"
Level 7	Decision 1	Choose whether to encrypt	If using public Wi-Fi to upload sensitive data, it's critical to encrypt the data first. "You're taking a risk by uploading sensitive data over someone else's Wi-Fi. What can you do to add a layer of protection?"

	Decision 2	Choose whether to encrypt	If using public Wi-Fi to upload sensitive data, it's critical to encrypt the data first. "You're taking a risk by uploading sensitive data over someone else's Wi-Fi. What can you do to add a layer of protection?"
Outcome 0 [Dead end]	Computer locks up		For students who reach Outcome 0: "As you try again, think about ways to ensure that you're using a secure network and that you're protecting your sensitive data. Also consider where you can secure your passwords to keep them safe and how you can stay focused on your work so you don't make mistakes as easily."
Outcome 1 [Sub-optimal]	Information is stolen		For students who reach Outcome 1: "As you try again, think about what kinds of links are safe to click, and consider how to keep information safe as it's transmitted across a public Wi-Fi connection."
Outcome 2 [Optimal]	Data is kept secure		For students who reach Outcome 2: "You got through the whole day without a security incident—good job!"

Module 7

Orientation		In this activity, students are challenged to make several decisions about how to develop a multimedia project. Along the way, they'll need to choose file types, software, delivery methods, and file locations. An important point to remember here is that some file types are better than others for specific situations, depending on the device(s) used, the delivery method, and the demands of the project (such as the need for speed vs. high quality). Students will also need to choose what software can best do needed tasks. Students should prioritize finding a good balance between high quality and low cost.	
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.
Level 1	Decision 1	Choose file format	WAV files tend to be very large to provide high quality. For streaming from an iPhone to wireless speakers, however, high quality is not so important. It's better to go for smaller files, even though the quality will be lower. MP3 and AAC are both good alternatives. "The goal here is to choose a file type that will meet the needs of the situation without being overly large in size. Which of these file types will work with an iPhone without taking up too much storage space?"
Level 2	Decision 1	Choose delivery method	The WAV file was too large for an email attachment and has to be delivered some other way. Using a USB flash drive for each of these files going forward would be time consuming and inconvenient. Optimally, students will use Dropbox. "What's the most convenient and fastest way to send each of these song files to the instructors?"
Level 3	Decision 1	Choose graphic source	Using licensed graphics is important from a legal standpoint, and finding images from a stock photo site provides a more consistent look. While it's also legally acceptable to draw original images, that will be more time consuming and might not look as professional in the end. "Keep in mind that you need a legal right to use the images you choose for these documents. How can you ensure that you get a professional and consistent look to your design?"
	Decision 2	Choose graphic source	Using licensed graphics is important from a legal standpoint, and finding images from a stock photo site provides a more consistent look. While it's also legally acceptable to draw original images, that will be more time consuming and might not look as professional in the end. "Keep in mind that you need a legal right to use the images you choose for these documents. How can you ensure that you get a professional and consistent look to your design?"
Level 4	Decision 1	Choose video editing app	Free apps can be a helpful tool, but in this case, students will need the more robust features offered by Camtasia. "How can you ensure you get the tools and features that you need in your video editing app?"

	Decision 2	Choose video editing app	Free apps can be a helpful tool, but in this case, students will need the more robust features offered by Camtasia. “How can you ensure you get the tools and features that you need in your video editing app?”
Level 5	Decision 1	Choose video resolution	Streaming videos from YouTube to a smartphone doesn’t require a high resolution. SD is sufficient for this task. “Which resolution will work best for streaming to smartphones?”
Level 6	Decision 1	Choose video streaming app	The purpose of the live video is to reach as many people as possible, and Skype isn’t designed for this purpose. Optimally, students will choose Facebook Live. “Which of these options will help you reach as many people as possible?”
	Decision 2	Choose video streaming app	The purpose of the live video is to reach as many people as possible, and Skype isn’t designed for this purpose. Optimally, students will choose Facebook Live. “Which of these options will help you reach as many people as possible?”
Level 7	Decision 1	Choose photo file format	JPEG files will provide the highest quality images for posting on social media, especially considering that many viewers will look at the photos on a large, desktop computer’s screen. “Lots of people look at social media photos on their larger computer screens, not just their smartphones. Which file type will ensure you’re showing the best quality photos?”
	Decision 2	Choose photo file format	JPEG files will provide the highest quality images for posting on social media, especially considering that many viewers will look at the photos on a large, desktop computer’s screen. “Lots of people look at social media photos on their larger computer screens, not just their smartphones. Which file type will ensure you’re showing the best quality photos?”
	Decision 3	Choose photo file format	JPEG files will provide the highest quality images for posting on social media, especially considering that many viewers will look at the photos on a large, desktop computer’s screen. “Lots of people look at social media photos on their larger computer screens, not just their smartphones. Which file type will ensure you’re showing the best quality photos?”
Outcome 0 [Dead end]	Must hire an IT consultant		For students who reach Outcome 0: “As you try again, think about how to invest sufficient resources to do a good job on this project without going overboard.”
Outcome 1 [Sub-optimal]	Mixed program results		For students who reach Outcome 1:

		“As you try again, think about how to balance quality with cost and convenience. Consider which device will be accessing each kind of resource, and look for ways to reach as many people as possible.”
Outcome 2 [Optimal]	Program is a resounding success	For students who reach Outcome 2: “The program has been a resounding success and it’s really making a positive impact on your community. Good job!”

Module 8

Orientation		In this activity, students are challenged to make decisions for developing an application for mobile devices. Along the way, they'll need to make design decisions, choose features and compatibility, and decide who to talk to on the development team to accomplish certain tasks. An important point to remember here is that a mobile app should be simple with a clean interface while still offering appealing interactions that bring users back to the app repeatedly. Students should prioritize a clean and engaging interface that is available to as many users as possible.		
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.	
Level 1	Decision 1	Choose app type	Students must choose between developing a mobile app to be installed on smartphones or developing a web app that is accessed through a web browser. Either option will meet the needs of the scenario. "Mobile devices use installed apps and web apps. Which kind of web do you want to develop?"	
Level 2	Decision 1	Choose OS platform	Students who chose to create an installed app must choose whether to develop an app that will work for only one major mobile OS (Android or iOS) or both. Optimally, students will choose to make the app compatible with both. "Which option will make your app attractive to as many users as possible?"	
Level 3	Decision 1	Choose fee	Students must decide whether to charge a small fee for the app or make it available for free. Optimally, the app will be free. "Remember, your goal is to attract as many potential customers as possible while ensuring they see value in your app. Do you think it's better to charge a few dollars for the app or make it free?"	
	Decision 2	Choose fee	Students must decide whether to charge a small fee for the app or make it available for free. Optimally, the app will be free. "Remember, your goal is to attract as many potential customers as possible while ensuring they see value in your app. Do you think it's better to charge a few dollars for the app or make it free?"	
	Decision 3	Choose fee	Students must decide whether to charge a small fee for the app or make it available for free. Optimally, the app will be free. "Remember, your goal is to attract as many potential customers as possible while ensuring they see value in your app. Do you think it's better to charge a few dollars for the app or make it free?"	
Level 4	Decision 1	Choose access type	Allowing anonymous users of the app limits the types of features that can be included but also makes for a simpler app design, fewer resources needed to support the app, and more privacy for users. Optimally for this track, students will choose to require users to create personalized accounts. This makes more sense for a paid app.	

			“If your users are paying for the app, they will expect higher end features. How can you ensure that your app can support all these interactive features?”
	Decision 2	Choose access type	Allowing anonymous users of the app limits the types of features that can be included but also makes for a simpler app design, fewer resources needed to support the app, and more privacy for users. Optimally for this track, students will choose to allow anonymous users. This makes more sense for a free app. “If your app is available for free, users might initially be suspicious of sharing their contact information. How can you invite users to try out the app without taking unnecessary risks with their information?”
Level 5	Decision 1	Choose features	Users like interactive features that give helpful information on the things they care about. By allowing a rating system and comments section, the app will support communication between users, which increases interactivity with the app. “How can you make the information available from this app increasingly valuable for users?”
	Decision 2	Choose features	Users like interactive features that give helpful information on the things they care about. By allowing a rating system and comments section, the app will support communication between users, which increases interactivity with the app. “How can you make the information available from this app increasingly valuable for users?”
Level 6	Decision 1	Choose graphics quality	Graphics quality can make or break an app. Simple and mismatched graphics give a shoddy appearance. Stock graphics will give a uniform, professional appearance, and custom graphics can further increase the app’s visual appeal. “What kind of graphics will ensure your app looks clean and professional?”
Level 7	Decision 1	Choose contractor	When needing to get an issue resolved, it’s important to contact the right person responsible for that work. In this case, the person who is in charge of coding the rating system is the programmer, not the graphics designer. “Make sure you contact the right person to address this issue. Who developed the rating system?”
	Decision 2	Choose contractor	When needing to get an issue resolved, it’s important to contact the right person responsible for that work. In this case, the person who is in charge of coding the rating system is the programmer, not the graphics designer. “Make sure you contact the right person to address this issue. Who developed the rating system?”
Outcome 0 [Dead end]		Project is canceled	For students who reach Outcome 0: “As you try again, think about how to make the app look clean and professional, what features are needed for it to be well received, and who is in charge of which parts of the project.”

Outcome 1 [Sub-optimal]	People don't like the app	For students who reach Outcome 1: "As you try again, think about who is in charge of which parts of the project and what features are needed for it to be well received by your target audience."
Outcome 2 [Optimal]	The app is hugely popular	For students who reach Outcome 2: "The app is a big hit, and your business' bottom line immediately starts to see positive results!"

Module 9

Orientation		In this activity, students are challenged to make decisions about how to update a website. Along the way, they'll need to decide how to interact with other people on the project, what features to include, how to approach the redesign, and how to fix problems. An important point to remember here is that this website belongs to an investment company, which serves experienced customers who expect professionalism and effectiveness. Students will need to choose a design that will appeal to these customers. Students should prioritize a sophisticated look and effective features.	
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.
Level 1	Decision 1	Choose renovation approach	Students need to choose whether to do the update themselves or to hire a website designer. Either approach is acceptable for this project. "The goal here is to update this website with a clean and professional design that will attract new customers to the investment company. How do you want to approach this project?"
Level 2	Decision 1	Choose level of detail in proposal	Even professional website designers need thorough information from their customers about what's needed for a website. In this case, more information is better. "How can you ensure that the website redesign accomplishes your goals?"
	Decision 2	Choose DIY tool	While it might take a little while to learn how to use it, a website builder is much more user friendly than trying to configure significant updates in Word using HTML. "What's the best tool to use for this website update? Remember that you need a clean and professional look when you're finished."
Level 3	Decision 1	Choose feature types	The purpose of this update is to attract new customers to the site. Videos added to the site might help existing customers but wouldn't attract new customers. A blog with posts on social media sites, some SEO work, and more efficient navigation features will bring new traffic to the website and provide a better user experience so, hopefully, some of that traffic will convert to new business. "What's the best way to get new visitors to the site?"
	Decision 2	Choose feature types	The purpose of this update is to attract new customers to the site. Videos added to the site might help existing customers but wouldn't attract new customers. A blog with posts on social media sites, some SEO work, and more efficient navigation features will bring new traffic to the website and provide a better user experience so, hopefully, some of that traffic will convert to new business. "What's the best way to get new visitors to the site?"
	Decision 3	Choose feature types	The purpose of this update is to attract new customers to the site. An updated logo, text, and links might help existing customers but wouldn't attract new customers. A blog with posts on social media sites, some SEO work, and more efficient navigation features will

			bring new traffic to the website and provide a better user experience so, hopefully, some of that traffic will convert to new business. “What’s the best way to get new visitors to the site?”
Level 4	Decision 1	Choose redesign strategy	The previous website was simple and lacking in quality, attractive design. So part of this update is to overhaul the layout and visuals. Starting from a professionally designed template can give the design process a boost, and then the template can be customized for specific needs. “How can you quickly get a working design for your website update, and then fine-tune it for your needs?”
Level 5	Decision 1	Choose design style	This website belongs to an investment company, which serves a clientele that expects a clean and sophisticated design. “What’s the best design style to attract customers to an investment company?”
	Decision 2	Choose design style	This website belongs to an investment company, which serves a clientele that expects a modern and sophisticated design. “What’s the best design style to attract customers to an investment company?”
Level 6	Decision 1	Choose which component to fix	Text can be difficult to read if there’s not much contrast between the text color and the background color. The website background needs to be edited so the text stands out better. “If there’s dark text on a dark page, it’s hard to read the text. What would need to change so there’s more contrast to see the text better?”
	Decision 2	Choose which component to fix	Text can be difficult to read if there’s not much contrast between the text color and the background color. The website background needs to be edited so the text stands out better. “If there’s dark text on a dark page, it’s hard to read the text. What would need to change so there’s more contrast to see the text better?”
Level 7	Decision 1	Choose tag	To add space between paragraphs, the <p> tags would format each paragraph in its own, separate space on the page. Adding tags between each paragraph would accomplish essentially the same goal by adding a blank line between paragraphs. “Which tags will add blank lines between paragraphs without affecting other formatting configurations?”
	Decision 2	Choose tag	To add space between paragraphs, the <p> tags would format each paragraph in its own, separate space on the page. Adding tags between each paragraph would accomplish essentially the same goal by adding a blank line between paragraphs. “Which tags will add blank lines between paragraphs without affecting other formatting configurations?”

Outcome 0 [Dead end]	Project is canceled	For students who reach Outcome 0: "As you try again, think about what kinds of features would most appeal to the intended audience of this website, which are customers of the investment firm."
Outcome 1 [Sub-optimal]	Owners aren't happy but move forward anyway	For students who reach Outcome 1: "As you try again, think about what features and tools are most appropriate along the way. Also be careful of the technical details when making changes."
Outcome 2 [Optimal]	Owners love the proposal	For students who reach Outcome 2: "The owners love the update—nicely done!"

Module 10

Orientation		In this activity, students are challenged to design a secure and functioning network. Along the way, they'll need to decide how to separate some types of traffic from other types of network traffic. An important point to remember here is what kinds of hardware the client needs and what kinds of configurations will accomplish the needed tasks at each step along the way. Students will also need to communicate with the client's customers in the face of a security threat. Students should prioritize security above convenience.		
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.	
Level 1	Decision 1	Choose package	Students are given two network device packages to choose from. Optimally, they will choose the smaller package, which is a better fit for the client's needs and budget. "The goal here is to choose a network device package that will service the client's needs without overselling. Which of these packages seems to be the best fit keeping in mind the client's requirements and budget?"	
Level 2	Decision 1	Concerned about customers	The client needs one portion of the network for customers and another secure portion for the cash registers and transaction traffic. Of the two options given, only the hardware firewall will provide sufficient security. "How can you provide sufficient protection of traffic from the cash registers so customers can't access that data?"	
	Decision 2	Concerned about customers	The client needs one portion of the network for customers and another, secure portion for the cash registers and transaction traffic. Of the two options given, neither will provide sufficient security. "How can you provide sufficient protection of traffic from the cash registers so customers can't access that data?"	
Level 3	Decision 1	Other stores	The client has additional store locations that need to share resources and data access. While either option here could be made to work, the cloud-based Intranet will provide a much more streamlined and fluid experience. "Which of these options will give all the stores the easiest access to data and other resources at different locations?"	
	Decision 2	Other stores	The client has additional store locations that need to share resources and data access. While either option here could be made to work, the cloud-based Intranet will provide a much more streamlined and fluid experience. "Which of these options will give all the stores the easiest access to data and other resources at different locations?"	
Level 4	Decision 1	Concerned about customer security	Security and convenience are often at odds with each other. In this case, the inconvenience of a complex password offers customers increased security, which is the client's top priority.	

			“Security and convenience are often at odds with each other. What is the customer’s top priority here? And how can you ensure that priority is met?”
	Decision 2	Concerned about customer security	Students on this track chose to use a VPN to secure one portion of the network from another. Neither option here will solve that problem. “What elements are needed for a VPN to work?”
	Decision 3	Concerned about customer security	Security and convenience are often at odds with each other. In this case, the inconvenience of a complex password offers customers increased security, which is the client’s top priority. “Security and convenience are often at odds with each other. What is the customer’s top priority here? And how can you ensure that priority is met?”
Level 5	Decision 1	Customers using internet	In this step, customers are accessing the Internet through an imposter’s Wi-Fi hotspot, which is likely a hacker. Optimally, students will warn customers not to use that network rather than signing into it themselves. “If this isn’t the store’s official network, is it safe? What should you do about that?”
	Decision 2	Customers using internet	In this step, customers are accessing the Internet through an imposter’s Wi-Fi hotspot, which is likely a hacker. Optimally, students will warn customers not to use that network rather than signing into it themselves. “If this isn’t the store’s official network, is it safe? What should you do about that?”
Level 6	Decision 1	Connect remotely with VPN	Here, students are able to use a VPN for the purpose it was designed: remote access. Students are challenged to include all the needed elements for the VPN to work, which includes both a server and a client component. “What elements are needed for a VPN to work?”
	Decision 2	Connect remotely with VPN	Here, students are able to use a VPN for the purpose it was designed: remote access. Students are challenged to include all the needed elements for the VPN to work, which includes both a server and a client component. “What elements are needed for a VPN to work?”
Outcome 0 [Dead end]		Nothing is working right	For students who reach Outcome 0: “As you try again, be careful to choose only enough hardware to meet the client’s needs. Also think about what kind of device can securely separate one portion of a network from another portion. Be careful to prioritize network security throughout.”
Outcome 1 [Sub-optimal]		Client is dissatisfied with the cost	For students who reach Outcome 1: “As you try again, be careful to choose only enough hardware to meet the client’s needs.”

Outcome 2
[Optimal]

Network is
functioning
well

For students who reach Outcome 2:

"The network is working well, and everything seems secure. Good job!"

Module 11

Orientation		In this activity, students are challenged to design a marketing strategy that connects with a target audience who is interested in drones and similar gadgets. Along the way, they'll need to decide how best to reach this audience. An important point to remember here is that tech savvy customers typically are comfortable with social media and other forms of digital communication. Students will also need to decide how to reach customers without compromising their privacy. Students should prioritize the use of digital communication, such as social media and email.	
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.
Level 1	Decision 1	Choose strategy	Students are given several options for designing a marketing strategy. Optimally, students will choose to research digital communication options. "The goal here is to use modern technology to reach your target audience. Is it better to rely on traditional techniques or try something new?"
Level 2	Decision 1	Choose tools for targeted audience	Social media ads and updates, in addition to blog posts and some new videos, can reach a wider audience for less money if the content is engaging and relevant to the target market. "Considering your tight budget, what's the best way to get the word out quickly without a high price tag?"
	Decision 2	Choose tools for modern market	Digital marketing techniques readily attract the attention of a tech savvy target market, especially when adding multimedia features such as videos and links to additional information. "Considering your tight budget and tech savvy audience, what kinds of marketing will attract the most attention and interaction from customers?"
Level 3	Decision 1	Choose microblogging frequency	Short posts don't take long to read, so frequent posts can keep the information fresh and current and people's minds. "Not every customer will see every post. So how can you increase the chances that customers will see multiple posts?"
Level 4	Decision 1	Choose hashtag	A hashtag should catch people's attention and also function as a search term to find additional information. By using a distinctive hash tag, it will better accomplish both these goals. "Which hashtag will catch people's attention and also make it easier to find your posts?"
	Decision 2	Choose hashtag	A hashtag should catch people's attention and also function as a search term to find additional information. By using a distinctive hash tag, it will better accomplish both these goals. "Which hashtag will catch people's attention and also make it easier to find your posts?"

Level 5	Decision 1	Choose profile page	Posts about new products should be posted primarily to the company's own page, which should already have several followers from their existing customer base. "Which profile page will likely have the largest following?"
	Decision 2	Choose profile page	Posts about new products should be posted primarily to the company's own page, which should already have several followers from their existing customer base. "Which profile page will likely have the largest following?"
Level 6	Decision 1	Choose email style	Emails from a company to their customers should have a professional tone that uses technological jargon appropriate to the audience. In this case, drone hobbyists will likely be interested in some of the tech details of the new product. "What kind of email message will best resonate with this company's existing customer base?"
	Decision 2	Choose email style	Emails from a company to their customers should have a professional tone that uses technological jargon appropriate to the audience. In this case, drone hobbyists will likely be interested in some of the tech details of the new product. "What kind of email message will best resonate with this company's existing customer base?"
Level 7	Decision 1	Choose email method	A BCC (blind carbon copy) will hide the email addresses of recipients, thus protecting customers' privacy. "How can you ensure that you protect your customers' privacy?"
	Decision 2	Choose email method	A BCC (blind carbon copy) will hide the email addresses of recipients, thus protecting customers' privacy. "How can you ensure that you protect your customers' privacy?"
Outcome 0 [Dead end]	Project given to someone else		For students who reach Outcome 0: "As you try again, think about how best to reach the kinds of people who would be interested in this product. Also consider how to protect their privacy when contacting them."
Outcome 1 [Sub-optimal]	Low attendance		For students who reach Outcome 1: "As you try again, consider that this company is communicating with its customers in a professional relationship. While it should be friendly and respectful, it should not be overly casual or personal."
Outcome 2 [Optimal]	Webinar is a success		For students who reach Outcome 2: "The webinar is a success, and the marketing campaign has brought in strong sales. Nice work!"

Module 12

Orientation		In this activity, students are challenged to make design decisions for a B2B website upgrade. Along the way, they'll need to decide how best to invest in modern technologies. An important point to remember here is that customers can feel insecure when faced with significant changes like this. Students will need to figure out how to make needed changes while avoiding any negative impact on how customers currently operate their own businesses. Students should prioritize wise investments in modern technology that will grow and adapt as the company succeeds.	
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.
Level 1	Decision 1	Choose cloud or on-prem web servers	Students are given the opportunity to choose cloud-host web servers or purchase new on-premises servers. Cloud servers are cheaper and offer more flexibility for growth and future improvements. "You have to replace the computer you've been using for your web server. Which of these options will give you the most flexibility for future growth and improvements?"
Level 2	Decision 1	Choose ecommerce platform	Here, students must choose an ecommerce platform type to incorporate into the new website. While an online marketplace might be cheaper, it won't offer the features needed for the scope of the website improvement project. "Which of these ecommerce platforms will give you the features you need for this extensive website upgrade?"
	Decision 2	Choose ecommerce platform	Here, students must choose an ecommerce platform type to incorporate into the new website. While an online marketplace might be cheaper, it won't offer the features needed for the scope of the website improvement project. "Which of these ecommerce platforms will give you the features you need for this extensive website upgrade?"
Level 3	Decision 1	Choose trust-building strategy	With such extensive upgrades to the online tools that will be available through the website, long term customers might feel insecure with the changes. The CEO has emphasized the need to help customers through this transition. Optimally, students will choose to increase available customer service hours. "Which strategy will best help customers navigate the planned changes to the website while feeling like their needs and expectations are being prioritized?"
	Decision 2	Choose trust-building strategy	With such extensive upgrades to the online tools that will be available through the website, long term customers might feel insecure with the changes. The CEO has emphasized the need to help customers through this transition. Optimally, students will choose to increase available customer service hours. "Which strategy will best help customers navigate the planned changes to the website while feeling like their needs and expectations are being prioritized?"

Level 4	Decision 1	Choose ecommerce job role	With increased customer service hours and newly available customer service tools on the website, students optimally will choose to hire more customer service representatives. “Considering the types of changes you’re proposing for the website, what job role is going to most need a boost in personnel?”
	Decision 2	Choose ecommerce job role	With increased customer service hours and newly available customer service tools on the website, students optimally will choose to hire more customer service representatives. “Considering the types of changes you’re proposing for the website, what job role is going to most need a boost in personnel?”
Level 5	Decision 1	Choose cloud service model	A SaaS CRM does not require any configuration or management of underlying servers. This kind of solution will also most likely offer built-in website integration tools. “Do you think it would be easiest to install CRM software on web-based servers using IaaS services, or would it be better to use SaaS CRM software that is accessed directly through the Internet?”
	Decision 2	Choose cloud service model	A SaaS CRM does not require any configuration or management of underlying servers. This kind of solution will also most likely offer built-in website integration tools. “Do you think it would be easiest to install CRM software on web-based servers using IaaS services, or would it be better to use SaaS CRM software that is accessed directly through the Internet?”
Level 6	Decision 1	Choose chatbot or live person	Either option here is a valid method of contacting customer service. Each option offers benefits for specific situations and needs. However, the chatbot offers the benefit of immediate availability. “Would you prefer to talk with a live person, or is it easier to chat with the chatbot?”
	Decision 2	Choose chatbot or live person	Either option here is a valid method of contacting customer service. Each option offers benefits for specific situations and needs. However, the chatbot offers the benefit of immediate availability. “Would you prefer to talk with a live person, or is it easier to chat with the chatbot?”
Level 7	Decision 1	Choose AI command	Chatbots are AI-driven and work best with clear and efficient requests for information that don’t include unnecessary background details. “Chatbots aren’t actually very chatty. What’s the best way to focus on the information you need from the chatbot?”
	Decision 2	Choose AI command	Chatbots are AI-driven and work best with clear and efficient requests for information that don’t include unnecessary background details. “Chatbots aren’t actually very chatty. What’s the best way to focus on the information you need from the chatbot?”

Outcome 1 [Optimal]	CEO is thrilled	For students who reach Outcome 1: “Great job designing a forward-thinking, well-organized, and cost-effective proposal!”
Outcome 2 [Sub-optimal]	Proposal is disorganized	For students who reach Outcome 2: “As you try again, think about how to use modern technology to help your company update its web-based services and best meet customers’ needs. Customers can feel insecure with these kinds of changes, so it’s important to consider how best to support them.”
Outcome 0 [Dead end]	Project is given to someone else	For students who reach Outcome 3: “As you try again, think about how to make wise investments in new technology that will best meet customer needs. Don’t be afraid of spending money that will help the business succeed.”

Module 13

Orientation		In this activity, students are challenged to make database design decisions for a new table. Along the way, they'll need to decide how to configure new fields. An important point to remember here is that a field name should reflect the data that field contains, and a field's data type should match the purpose of the data in the field. Students will also need to choose a relationship type between tables that shows how records in one table relate to records in the other table. Students should prioritize database structures that help ensure users will enter accurate information.	
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.
Level 1	Decision 1	Choose primary key	Students are asked to identify which field will best serve as the table's primary key. Students need to recognize that a primary key field should always contain unique data for each record. Because many people can share the same name, a Client Name field won't work as a primary key. "Which of these fields will always hold a unique piece of information for every client?"
Level 2	Decision 1	Choose field name	Here, students are tempted to name the primary key field PrimaryKey. However, fields should always be named for the kind of data they hold, not the purpose they serve in the database. "Which of these field names will be most informative if database users see the field name in queries and reports?"
	Decision 2	Choose field name	Here, students are tempted to name the primary key field PrimaryKey. However, fields should always be named for the kind of data they hold, not the purpose they serve in the database. "Which of these field names will be most informative if database users see the field name in queries and reports?"
Level 3	Decision 1	Choose data type	While the Number data type is very tempting, the phone number field will not be used for calculations. The module's narrative gives this exact example of a phone number field for when to use a Short Text data type on a field that contains numbers. "The Number data type is typically used for fields that will be included in calculations. Is that data type appropriate here?"
	Decision 2	Choose data type	While the AutoNumber data type is tempting, the phone number field should not be automatically generated. "The AutoNumber data type automatically generates a number for each new record. Is that data type appropriate here?"
	Decision 3	Choose data type	While the AutoNumber data type is tempting, the phone number field should not be automatically generated. "The AutoNumber data type automatically generates a number for each new record. Is that data type appropriate here?"

Level 4	Decision 1	Choose validity check	Here, students are asked to identify what kind of validity check will best assist database users in confirming the phone number for a new record entry. Phone numbers are not always unique (such as when two people share the same work number). But phone numbers do conform to a specific format, which can be enforced with an input mask. “Which of these checks will remind users to input exactly the right number of digits for a phone number?”
	Decision 2	Choose validity check	Here, students are asked to identify what kind of validity check will best assist database users in confirming the phone number for a new record entry. Phone numbers are not always unique (such as when two people share the same work number). But phone numbers do conform to a specific format, which can be enforced with an input mask. “Which of these checks will remind users to input exactly the right number of digits for a phone number?”
	Decision 3	Choose validity check	Here, students are asked to identify what kind of validity check will best assist database users in confirming the phone number for a new record entry. Phone numbers are not always unique (such as when two people share the same work number). But phone numbers do conform to a specific format, which can be enforced with an input mask. “Which of these checks will remind users to input exactly the right number of digits for a phone number?”
Level 5	Decision 1	Choose relationship type	The one-to-many relationship shows that each agent can support many clients. A one-to-one relationship would require that each agent have only one client. The many-to-many relationship would allow each client to have multiple agents as well as each agent supporting multiple clients. While this is theoretically possible, it’s not a likely scenario. “Ask yourself two questions: Can each agent have more than one client? Will each client be assigned to more than one agent at this agency? The answer to these questions can help you evaluate which type of relationship you need between these two tables.”
	Decision 2	Choose relationship type	The one-to-many relationship shows that each agent can support many clients. A one-to-one relationship would require that each agent have only one client. The many-to-many relationship would allow each client to have multiple agents as well as each agent supporting multiple clients. While this is theoretically possible, it’s not a likely scenario. “Ask yourself two questions: Can each agent have more than one client? Will each client be assigned to more than one agent at this agency? The answer to these questions can help you evaluate which type of relationship you need between these two tables.”
Level 6	Decision 1	Choose query parameter	This decision requires students to choose between two basic mathematical statements. Clients born after 1990 will have a birth year that is greater than 1990 (such as 1991, 1992, etc). Therefore, the > (greater than) symbol is required for the query. “Think of some birth years for clients who were born after 1990. How do those years compare in value to 1990? Then decide which of these mathematical statements would give you those kinds of results.”

	Decision 2	Choose query parameter	This decision requires students to choose between two basic mathematical statements. Clients born after 1990 will have a birth year that is greater than 1990 (such as 1991, 1992, etc). Therefore, the > (greater than) symbol is required for the query. “Think of some birth years for clients who were born after 1990. How do those years compare in value to 1990? Then decide which of these mathematical statements would give you those kinds of results.”
Outcome 1 [Optimal]	Clients table works well		For students who reach Outcome 1: “Nice work designing this new table! The real estate agents are pleased, and Clarence’s business is attracting more interest.”
Outcome 2 [Sub-optimal]	Clients table is disorganized		For students who reach Outcome 2: “As you try again, think about which data type is best for each new field, especially for a number field that is not used in calculations. Also consider what kind validity check will work best for that phone number field.”
Outcome 0 [Dead end]	Clarence hires a consultant		For students who reach Outcome 0: “As you try again, think about what information users need to see as they’re working with fields in the database. You can use field names to give insight on what a field’s purpose is. Also consider which data type is best and how to assist users in providing accurate information.”

Module 14

Orientation		In this activity, students are challenged to make ethical decisions responsibly. Along the way, they'll need to decide how to consider laws and rules that should guide their behavior. An important point to remember here is that most decisions have consequences and could be discovered by other people. Students will need to consider what those consequences could be. Students should prioritize making responsible choices that preserve safety, integrity, and respect for others.	
Level	Decision	Description	Strategy: Comments you can give your students to help them think through each decision point.
Level 1	Decision 1	Choose how to use info source	Students are asked to decide how to use an information source in a way that will not result in plagiarism. While it's fine to use someone else's ideas as a starting point, changing a few words is not enough to prevent plagiarism. When writing an article, the article should consist of the author's own ideas. "How can you make sure that you don't plagiarize someone else's words or ideas? Think about the steps you need to take to ensure the article contains your own ideas."
Level 2	Decision 1	Choose device for personal message	Most companies have a clause in their AUP (acceptable use policy) that states company computers can only be used for work purposes. While there are exceptions, personal messaging should still only occur during break times. "It's often helpful to ask yourself, 'How would you handle this situation if your boss were standing right behind you?'"
Level 3	Decision 1	Choose response to complaint	One never knows who else might see an email conversation. Anything communicated digitally should take into consideration possible consequences if the emails were shared publicly. It's also crucial to not speak disparagingly of a group of people. This could be considered discrimination or hate speech and could incur legal consequences. "How can you help your coworker while also showing respect for the customer's need for more information?"
	Decision 2	Choose response to complaint	One never knows who else might see an email conversation. Anything communicated digitally should take into consideration possible consequences if the emails were shared publicly. It's also crucial to not speak disparagingly of a group of people. This could be considered discrimination or hate speech and could incur legal consequences. "How can you help your coworker while also showing respect for the customer's need for more information?"
Level 4	Decision 1	Choose post on social media	Like the email situation, it's important to consider what kind of impact comments on social media can have on one's reputation. "What kind of social media post will help build your reputation for being professional and reliable? Think about the topic and your approach to writing the post."

Level 5	Decision 1	Choose whether to use access to client info	It's unethical to use private client information for personal purposes. When handling a personal situation, it's always more professional to go through legitimate channels to obtain information. "Consider how you would feel if a neighbor who was angry at you used their role at work to access your information as a customer at their store. Would this help you work with them to solve the problem, or would it make the problem between you worse?"
Level 6	Decision 1	Choose whether to keep driving	Most states have distracted driving laws prohibiting the use of a phone while driving if the phone must be held by the driver. Even in the absence of such laws, an important video call such as this one with a client deserves full attention. "How can you make sure you offer your client your best work? It's also important to think about your safety with the use of technology."
	Decision 2	Choose whether to keep driving	Most states have distracted driving laws prohibiting the use of a phone while driving if the phone must be held by the driver. Even in the absence of such laws, an important video call such as this one with a client deserves full attention. "How can you make sure you offer your client your best work? It's also important to think about your safety with the use of technology."
Level 7	Decision 1	Choose how to dispose of old device	Electronic waste has a significantly negative impact on the environment. However, recycled and refurbished electronics can help accomplish helpful purposes. "Is an old phone worth anything to anyone else? Sometimes old phones can be refurbished and then used by someone else. Also, think about the environmental impact if you were to not properly dispose of your phone, which contains hazardous chemicals that are dangerous to the environment."
	Decision 2	Choose how to dispose of old device	Electronic waste has a significantly negative impact on the environment. However, recycled and refurbished electronics can help accomplish helpful purposes. "Is an old phone worth anything to anyone else? Sometimes old phones can be refurbished and then used by someone else. Also, think about the environmental impact if you were to not properly dispose of your phone, which contains hazardous chemicals that are dangerous to the environment."
Outcome 1 [Optimal]	Earn promotion	For students who reach Outcome 1: "Good work getting the promotion and a couple of fun vacation days—making hard decisions can really pay off!"	
Outcome 2 [Sub-optimal]	Need more training	For students who reach Outcome 2: "As you try again, think about how to make professional and responsible decisions even when it's tempting to cut corners or think that no one will find out. Also keep in mind that, while technology can greatly improve your life, it also requires a level of responsibility to ensure you use tech safely."	

Outcome 0 [Dead end]	At risk of losing job	For students who reach Outcome 0: “As you try again, consider what laws or guidelines might apply to the decisions you need to make. How can you ensure that you abide by the intent of the law, not just seeing what you can get away with?”
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